Process scheduling

1) Three processes arrive at time 0

process	Burst time
P_1	5
P_2	3
P_3	8

using FCFS, the average waiting time is:

Average waiting time = $\frac{\text{Total waiting time}}{\text{No. of processes}}$

using FCFS, processes execute in the order $P_1 \rightarrow P_2 \rightarrow P_3$

P_1 waiting time = 0

P_2 waiting time = 5

P_3 waiting time = 5+3 = 8

Average wt = $\frac{0+5+8}{3} = \frac{13}{3} = 4.33$

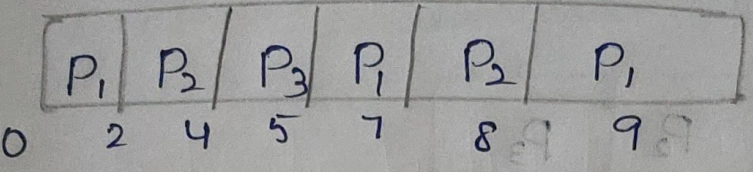
2) Consider Round Robin with time quantum 2

process	Burst time
P_1	5
P_2	3

All arrive at time 0. which process completes first?

Each process gets a maximum of 2 time units per turn

Gantt chart



P_1 executes from 0-2; remaining bt is 3

P_2 executes from 2-4; remaining Bt is 1

P_3 executes from 4-5, completens than P_2 and P_1

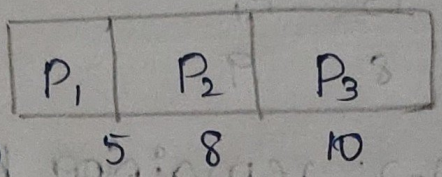
P_3 completes first

3) consider the following process scheduled using FCFS, All process arrive at time 0

process	Burst time
P_1	6ms
P_2	2ms
P_3	8ms
P_4	3ms

Process	AT	BT
P_1	0ms	5ms
P_2	2ms	3ms
P_3	4ms	2ms

$P_1 \rightarrow P_2 \rightarrow P_3$



P_1: CT = 0+5 = 5ms

P_2: CT = 5+3 = 8ms

P_3: CT = 8+2 = 10ms

TAT = CT − AT

Process	CT	AT	TAT
P_1	5	0	5−0=5
P_2	8	2	8−2=6
P_3	10	4	10−4=6

Total = 17ms

Average TAT = $\frac{\text{Total TAT}}{\text{No: of process}} = 17/3$

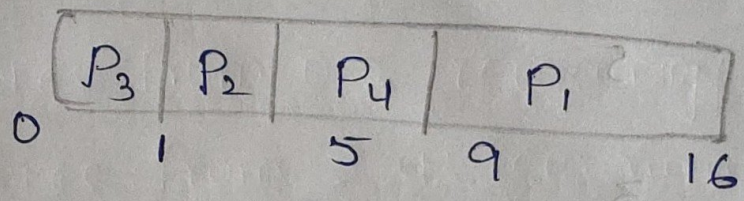
= 5.67ms

4)

process	Burst time
P_1	7ms
P_2	4ms
P_3	1ms
P_4	4ms

what is average waiting time,

Execution order: $P_3 \longrightarrow P_2 \longrightarrow P_4 \longrightarrow P_1$



$P_3 \rightarrow 0$

$P_2 \rightarrow 1$

$P_4 \rightarrow 5$

$P_1 \rightarrow 9$

Total = 15

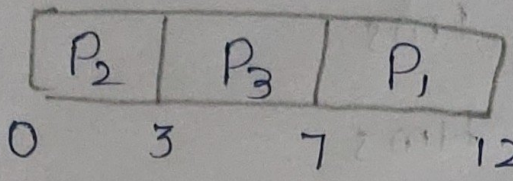
Average wt = $\frac{15}{4}$ = 3.75 ms

5) All process arrive at time 0

Process	BT	priority
P_1	5ms	3
P_2	3ms	1
P_3	4ms	2

What is average waiting time using non premptive priority scheduling.

Execution order :- $P_2 \rightarrow P_3 \rightarrow P_1$



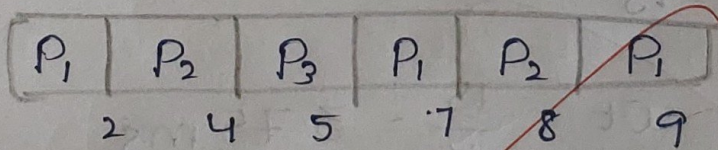
Average wt $= \frac{0+3+7}{3} = 3.33$ ms

6)

Process	BT
P_1	5
P_2	3
P_3	1

What is the completion time of P_2

Time quantum = 2ms



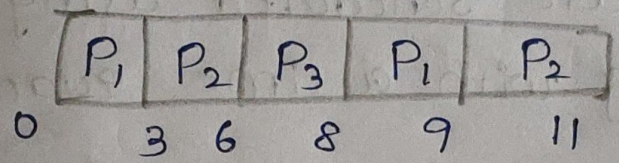
Process	completion time
P_1	9ms
P_2	8ms
P_3	5ms

completion time of P_3 = 8ms

7) Round Robin with a time quantum of 3ms

process	Burst time
P_1	4 ms
P_2	5ms
P_3	2ms

Time Quantum = 3ms

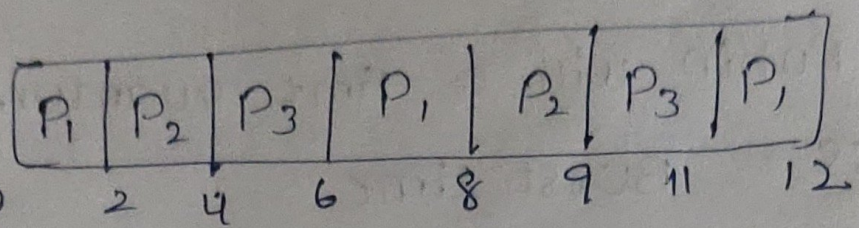


process	completion Time
P_1	9ms
P_2	11ms
P_3	8ms

8) Time quantum = 2ms

process	Arrival time	Burst time
P_1	0	5
P_2	1	3
P_3	2	4

Gantt chart



P_1 = 12 ms

P_2 = 9 ms

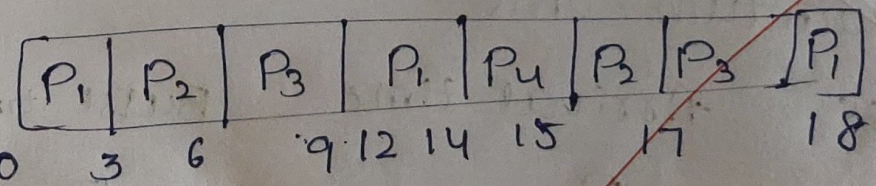
P_3 = 11 ms

completion time of P_2 = 9 ms

9) Four processes are schedoled using roun robin scheduling with a time quantum 3 ms

process	AT	BT
P_1	0	7
P_2	2	4
P_3	3	5
P_4	5	2

Gantt chart



wt = CT - AT - BT

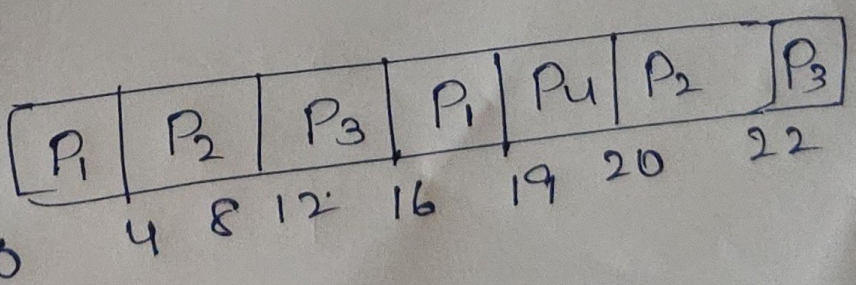
process	AT	BT	CT	waiting time
P_1	0	7	18	18-0-7 = 11
P_2	2	4	15	15-2-4 = 9
P_3	3	5	17	17-3-5 = 9
P_4	5	2	6	6-5-2 = 1

Average waiting time = $\frac{11+9+9+7}{4}$

$= \frac{36}{4} = 9$ ~~mins~~ 8.25 m

10) Find the Average turnaround time Four processes arrives at different times and are executed using round robin scheduling with a time quantum of 4ms

process	Arrival time	Burst time
P_1	0	8
P_2	1	5
P_3	3	6
P_4	6	3



TAT = CT − AT

Process	AT	BT	CT	TAT
P_1	0	8	16	16 - 0 = 16
P_2	1	5	20	20 - 1 = 19
P_3	3	6	22	22 - 3 = 19
P_4	6	3	19	19 - 6 = 13

$$\text{Average TAT} = \frac{16 + 19 + 19 + 13}{4}$$

$$= \frac{67}{4} = 16.75 \text{ ms}$$

Good
14/10/20

